期末测验

2019年1月7日 21:23

1 单选 (2分) In the object-oriented system, we use_____relationship to express that a larger “whole” class includes one or more smaller “partial” classes.

- A. generalization
- B. aggregation
- C. inheritance
- D. composition

2 单选 (2分) Within the object-oriented technology, objects are instances of classes; there are three components for an object:: _____、attribute and methods (or operations).

- A. rule
- B. identification
- C. encapsulation
- D. message

3 单选 (2分) When we choose an object-oriented programming language for software development, we don’t need to considerate whether this language_____.

- A. will take the leading position of the market
- B. has an abundant class library
- C. has an mature development environment
- D. supports definition of global variable and global functions

4 单选 (2分) Considering the various UML views, _____(1)_____shows the system function observed by external participants;_____(2)____shows how the system function is implemented internally from the point of the system’s static structure and dynamic behaviors；_____(3)_____shows the structures of source code and actual executable code.

- A. (1) usecase view (2) process view (3) logical view
- B. (1) usecase view (2) logical view (3) process view
- C. (1) logical view (2) process view (3) usecase view
- D. (1) usecase view (2) logical view (3) implementation view

5 单选 (2分) ____is a dataflow-oriented development method，the basic idea of which is the decomposition and abstraction of software functions.

- A. structured development method
- B. Jackson system development method
- C. Booch method
- D. UML（Unified Modeling Language）

6 单选 (2分) When using UML for software design, we can use ____relationship to express the special/general relation between two classes of affairs.

- A. dependency
- B. aggregation
- C. generalization
- D. implementation

7 单选 (2分) The main task of object-oriented analysis (OOA) is to _____

- A. build the logical model using objects as basic units
- B. make users’ information requirements clear
- C. determine the overall plan of the system
- D. identify objects together with their attributes and operations

8 单选 (2分) Which of the following is NOT correct? _____

- A. Object-oriented programming languages can support procedural programming.
- B. The time complexity of a given algorithm has no relation with the programming language

A. Object-oriented programming languages can support procedural programming.

B. The time complexity of a given algorithm has no relation with the programming language used to implement it.

C. Compared with assembly language, programming in script languages can obtain higher execution efficiency.

D. Object-oriented programming languages don’t support the direct access to a member variable of an object.

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9 单选 (2分) ___(1)_____is to compose data and methods for date operation into the same unit so that we can take classes as the basic reuse units, increasing its cohesion and decreasing its coupling. In the object-oriented method, _____(2)_____mechanism is the simulation of the inheritance in the real world, by this mechanism, the attributes and methods of the base class are inherit by the derived classes.

A. (1) encapsulation (2)inheritance

B. (1) encapsulation (2)polymophism

C. (1) inheritance (2)polymophism

D. (1) inheritance (2)encapsulation

10 单选 (2分) Message delivery is the communication method between objects. An object request services from another object by sending messages. A message usually includes: _____

- A. message-sender’s identification, operation name and necessary parameters
- B. the class names of message-sender and message-receiver
- C. message-receiver’s identification, operation name and necessary parameters
- D. the class name of the message-receiver

11 单选 (2分) _____is used to express the relation between a class and one or more detailed classes, i.e., the relation between “general” and “special”.

- A. generalization
- B. association
- C. aggregation
- D. composition

12 单选 (2分) The best tool for further modeling of business process flow in a complex case is _____

- A. State Diagram
- B. Sequence Diagram
- C. Class Diagram
- D. Activity Diagram

13 单选 (2分) Object - oriented analysis requires to find out all entities (concepts) objectively existing in the software requirements, and then summarize and abstract entity classes. _____ is a useful approach to find out entities.

A. Meeting Survey

B. Questionnaire

C. Telephone Survey

D. Noun Phrase Analysis

14 单选 (2分) Object-oriented analysis and design are two important phases in software development, _____ of the following doesn't belong to object –oriented analysis.

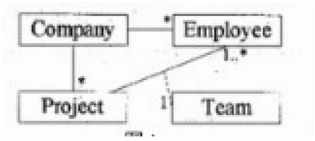
A. Construct analysis model

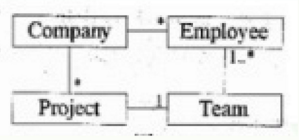
B. Identify analysis classes

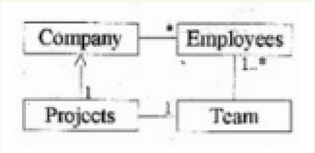
C. Determine interface specification

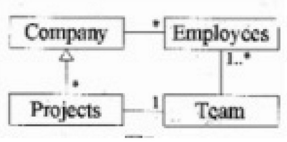
D. Evaluate analysis model

15 单选 (2分) A company take charge of several projects, and each project is developed by a employee team. _____ of the following UML concept diagram is most suitable for this scenario.

A. 

B. 

C. 

D. 

16 单选 (2分) There are following business statements for an information system:①A customer submit 0 or more orders;②An order can only be submitted by one customer. There are two classes in the system "Customer" and "Order". For each instance of Order Class, there are ____ instance(s) of Customer Class; while for each instance of Customer Class, there are ____ instance(s) of Order Class.

A. (1) 1 (2) 1

B. (1) 1 (2) 1 or more

C. (1) 1 (2) 0 or more

D. (1) 0 or more (2) 1

17 单选 (2分) In object-oriented technology, objects have the following characteristics:

（1）clear boundary（2）well-defined behavours（3）fixed position and quantity（4）Scalability

Which of the above are correct?

- A. （2）（4）
- B. （1）（2）（3）（4）
- C. （1）（2）（4）
- D. （1）（2）

18 单选 (2分) Which of the following statements concerning object-oriented analysis is wrong?

- A. Object-oriented analysis focuses on analyzing problem domain and system responsebilities.
- B. Object-oriented analysis needs to consider system testing.
- C. Object-oriented analysis igonores problems related with system implementation.
- D. Object-oriented analysis construct implementation-independent system analysis model

19 单选 (2分) Which of the following statements concerning object-oriented design is wrong?

- A. High-level modules should not depend on low-level modules.
- B. Abstractions should not depend on details.
- C. Details can depend on abstractions.
- D. High-level modules cannot be independent with low-level modules.

20 单选 (2分) Business usecase together with actors depict _____.

- A. static elements in the business process
- B. dynamic elements in the business process
- C. logic view of business process
- D. business process supported by organizations

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主观题:

1 （4分） What is Software Process Model? And what should be included in it?

2 （4分）

According to what is explained in class, how many traditional methods of requirements elicitation are there? What are they?

3 （4分）

According to what is explained in class, how many categories are there for UML models? Explain the functions of each model.

4 （4分）

When we talk about persistence and database design, how many levels of data models are there? What are they? Please explain your answer.

5 （4分）

According to what is explained in class, what should be included in a usecase model? What's the difference between usecase inclusion and usecase extension?

6 （20分）

A university has several departments. Each department has several classes and research institutes. Each research institute has several staff members, among which some professors and associated professors supervise some postgraduates. Each class has several students. Each student can opt several courses and each course can be opted by several students.

（1）Create corresponding concept model using noun/verb method.

（2）Convert the concept model from Question 1 to corresponding database model, described by tables.

7 （20分）

The main users of Stock Management System are Instock Administrators, Outstock Administrators, Storage Administrators and System Administrators. Instock administrators are responsible for the registry, update and query of instock journal. Outstock administrators are responsible for the registry, update and query of outstock journal. Storage Administrators are responsible for the registry of Stock ledger, the query of stock information and printing stock reports. System administrators are responsible for user management, data backup and product fundamental information management.

Create the system use case diagram and class diagram.